



THE POWER OF VACCINATION

From Smallpox Eradication to COVID-19
in Record Time

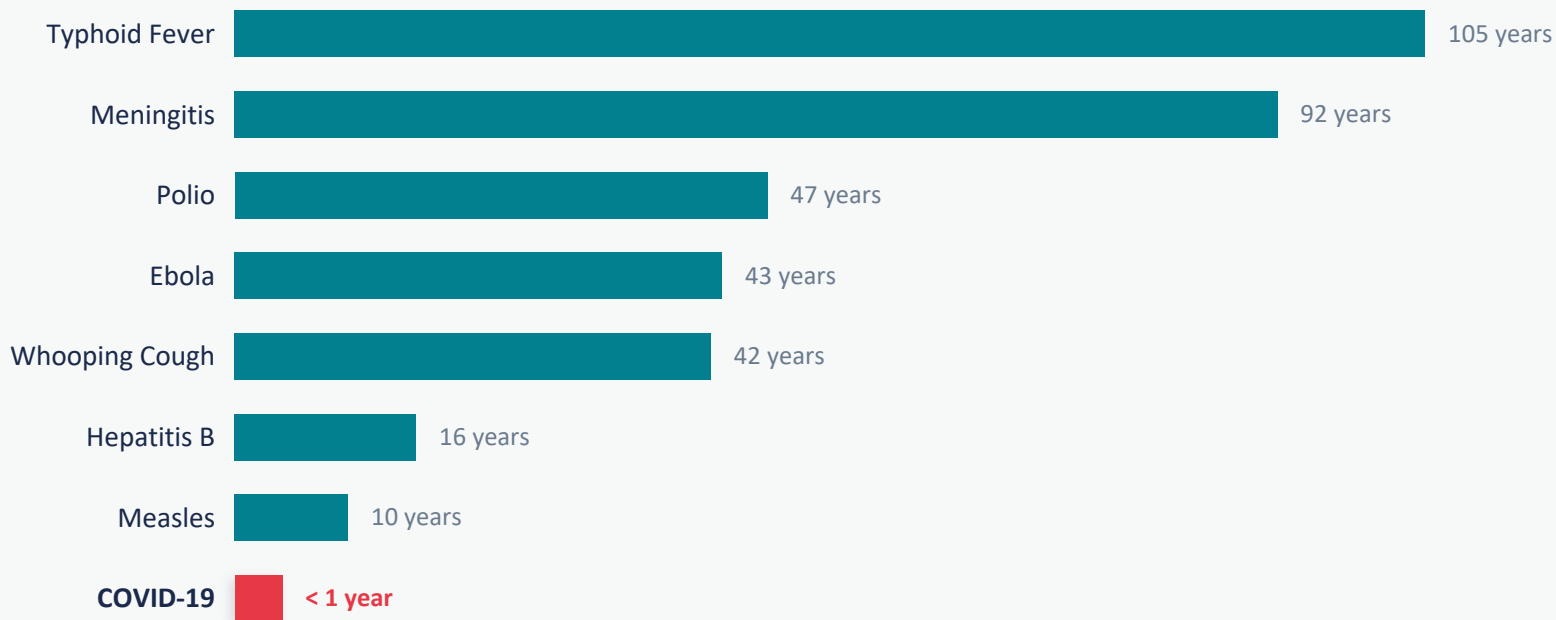
A Data-Driven Overview for Public Health Professionals & Policy Analysts

Spanning Two Centuries of Innovation: 1796 – 2020

Source: Our World in Data • ourworldindata.org/vaccination

From 100 Years to Under 1 Year

Vaccine Development Is Accelerating



COVID-19 vaccine developed in under 1 year — enabled by mRNA technology and genome sequencing

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox

Pre-vaccine vs. Post-vaccine Annual Rates in the United States (per million)

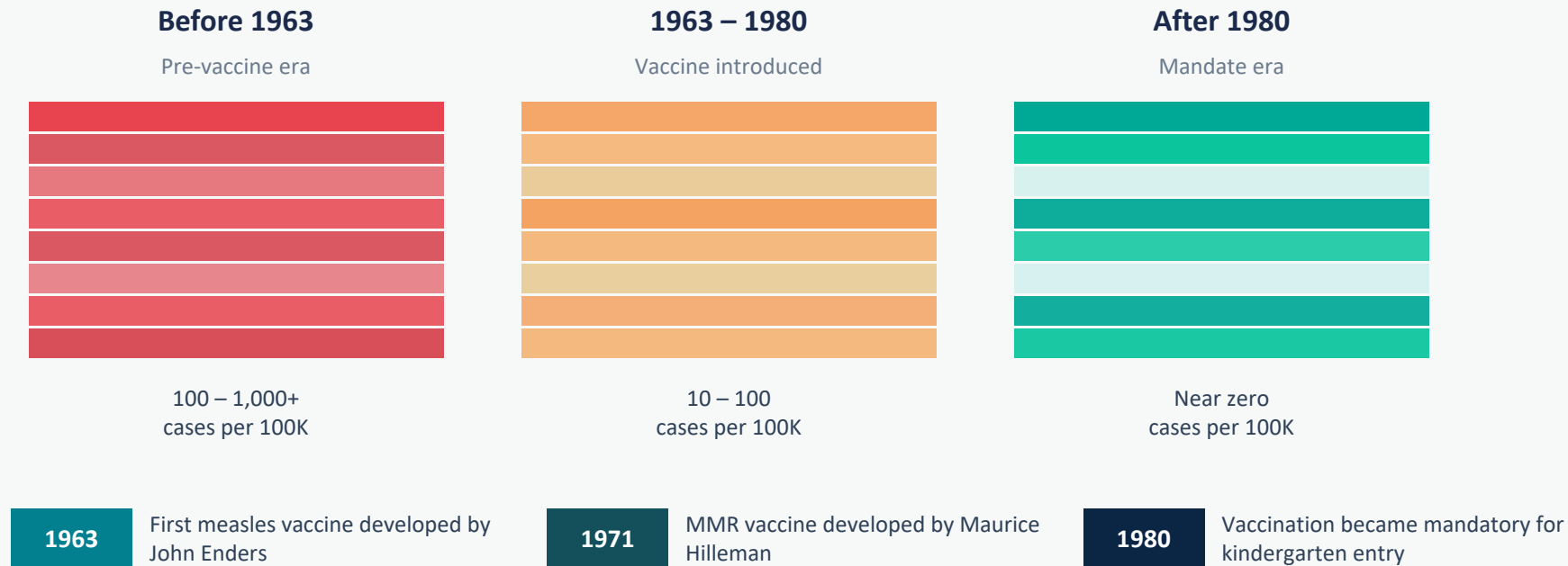
Disease	Pre-Vaccine Cases	Post-Vaccine Cases	Case Reduction	Pre-Vaccine Deaths	Death Reduction
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute Polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

Highlighted rows = 100% case or death elimination achieved

Source: Roush & Murphy (2007)

Measles Cases Collapsed Across All 50 US States

State-by-state measles incidence per 100,000 (1929–2022) with key policy milestones



Before 1963, nearly every state experienced high measles incidence. After the 1980 mandate, cases dropped to near zero with only sporadic outbreaks.

Smallpox: The Only Human Disease Eradicated Through Vaccination



Why Smallpox Was Uniquely Suited for Eradication

Human-Only Host

The virus infected only humans — no animal reservoir to sustain transmission

Distinguishable Symptoms

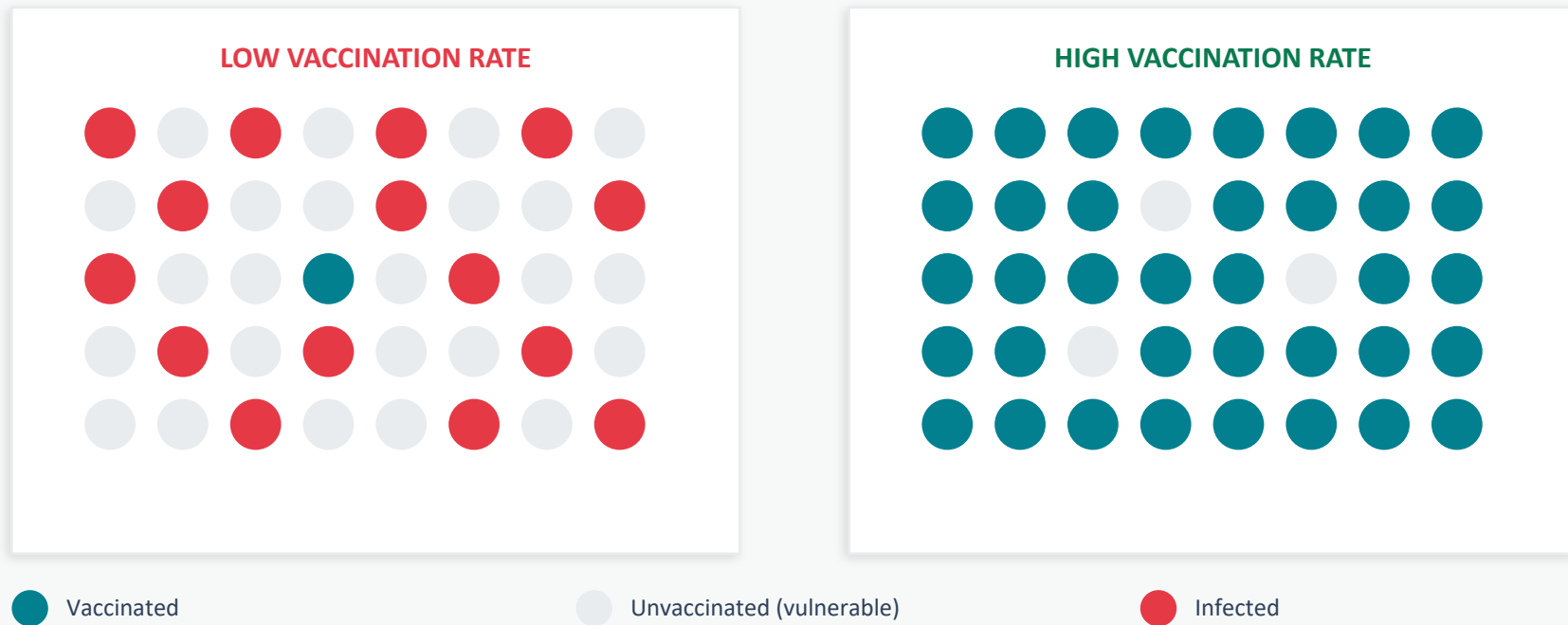
Clinical symptoms were clearly identifiable for rapid case detection

Effective & Cheap Vaccine

The vaccine was inexpensive, stable, and highly effective

250 cases per million/year → 0 | 2.9 deaths per million/year → 0 | 100% elimination

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable



When enough people are vaccinated, pathogens cannot spread effectively — shielding **newborns, immunocompromised individuals, and cancer patients on chemotherapy** who cannot be vaccinated themselves.

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Pneumococcal

40.5%

case reduction

31.3% death reduction

Hepatitis A

89%

case reduction

88.7% death reduction

Hepatitis B

83.9%

case reduction

83.6% death reduction

Key Barriers to Closing the Gap

Access Gaps

Not everyone has access to vaccines globally

Supply Shortages

Supply shortages put children at risk

Trust Challenges

Poor scientific understanding reduces trust and uptake

Compared to 100% elimination achieved for diphtheria, smallpox, and polio — these diseases demand continued investment.

Key Takeaways

Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done



Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and polio in the US



Development timelines collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology



Measles cases dropped to near zero across all 50 US states after the 1980 kindergarten vaccination mandate



Smallpox remains the only human disease eradicated through vaccination, certified in 1980



Gaps persist: pneumococcal disease (40.5%), hepatitis A (89%), and hepatitis B (83.9%) need improved coverage

CALL TO ACTION

Global health systems must improve vaccine access, close supply gaps, and build public trust to replicate these successes for the diseases where the job remains unfinished.